



Yama

Carbon dioxide removal prepurchase application 2025

General Application

(The General Application applies to everyone; all applicants should complete this)

Public section

The content in this section (answers to questions 1(a) - (d)) will be made public on the [Frontier GitHub repository](#) after we announce purchases in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Company or organization name

Yama

Company or organization location (we welcome applicants from anywhere in the world)

France, Paris

Name(s) of primary point(s) of contact for this application

Aurélie Gonzalez, co-founder & CEO

Brief company or organization description <20 words

France-based DAC startup using non-toxic and hybrid electrochemical - thermal tech to remove CO₂ at scale at < 150\$/tCO₂ with low energy consumption

1. Public summary of proposed project¹ to Frontier

- a. **Description of the CDR approach:** Describe how the proposed technology removes CO₂ from the atmosphere, including how the carbon is stored for > 1,000 years. Tell us why your system is best-in-class, and how you're differentiated from any other organization working on a similar approach. If your project addresses any of the priority innovation areas identified in the RFP, tell us how. Please include figures and system schematics and be specific, but concise. 1000-1500 words

¹ We use "project" throughout this template, but the term is not intended to denote a single facility. The "project" being proposed to Frontier could include multiple facilities/locations or potentially all the CDR activities of your company.

Yama is pioneering DAC 3.0: the next-generation direct air capture platform that merges the high energy efficiency and selectivity of electrochemical approaches (DAC 2.0) with the deployment speed and robustness of earlier DAC systems (DAC 1.0). Our hybrid process integrates fast kinetics chemistry solvent with bipolar electrodialysis (BPED) and low-grade heat. It uses a patent-protected, non-toxic solvent blend along with custom membrane stacks to achieve maximal performance, reliability, and modular scalability.

Process description and durability

At the heart of our technology is a proprietary process for capturing and concentrating atmospheric CO₂ at industry-leading energy and cost metrics. Ambient air is brought into contact with a circulating, water-based solution of carefully selected non toxic fast kinetics within a compact contactor. Our solvent is tuned for rapid reaction kinetics with CO₂, allowing for smaller equipment, low pressure drop, and rapid air throughput. Unlike classic amine, our blend is non-volatile and non-toxic, with negligible losses to degradation or evaporation—key for safe, urban, and large-scale deployment.

The CO₂-rich solution is then directed into our custom-designed BPED stack. Here, we decouple the absorption and desorption pH: using selective membranes and a carefully managed electrochemical environment, we react bicarbonate anions with protons generated at the membrane interface. This releases pure CO₂ in a highly controlled and energy-efficient manner. The proton release and bicarbonate separation processes are confined within the stack, protecting both solvent and membrane from degradation. By optimizing the pH swing and stack geometry, we significantly reduce bipolar membrane overpotential, allowing operation at <1800 kWh/tCO₂ for our next pilot (current project) and with a next-generation target of <1250 kWh/tCO₂.

Desorption is powered by low-grade heat, widely available from sources like waste heat from data centers or heat produced on purpose by heat pumps, and the closed-loop solvent system is continuously recycled. Our pilot units have demonstrated solvent durability and consistent performance, and our stack design—protected by two patents—ensures system longevity at scale.

After CO₂ release, the gas is transported on a short distance on-site and injected into cement/concrete waste fines. In this mineralization stage, the captured CO₂ reacts with residual calcium silicates and hydroxides to form solid calcium carbonates (CaCO₃), producing carbon-negative aggregates that are immediately reused as sustainable building materials. This mineralization is rapid, permanent, and requires no pipelines, deep-well injection, or hazardous waste handling. The resulting storage is geologically stable for >1,000 years, and MRV can be performed through direct mass balance and aggregate analysis.

Differentiation and best-in-class performance

Yama's DAC 3.0 platform addresses and overcomes the chronic limitations of both first-generation (thermal) and second-generation (electrochemical) DAC approaches. Our hybrid electrochemical solution stands out through:

- Unmatched energy efficiency: Our current system at prototype demonstrates <2000 kWh/tCO₂ from lab to pilot (TRL6). This leap is enabled by both solvent innovation and stack design, delivering best-in-class energy use for any large-scale DAC.
- Cost trajectory and system economics: Our process delivers a credible pathway to <\$350/tCO₂ at FOAK - in 2030, for a market ready to pay for high quality credits at around 300-400\$/tCO₂ at that time - and <\$150/tCO₂ at the million-ton scale. This is achieved by minimizing Capex (modular, containerized, off-the-shelf components) and Opex (solvent and

membrane durability, electricity, maintenance). In contrast, classic DAC 1.0 systems (thermal) remain above \$600/tCO₂ due to energy intensity, and many DAC 2.0 (other electrochemical) systems face cost and scale constraints from rare materials or risky solvents, unscalable, and claim 300-400\$/tCO₂ at million tons scale.

- Scalability and engineering: Unlike most electrochemical approaches, Yama's system is free from rare/exotic materials and does not depend on volatile, hazardous solvents. Our process uses mature supply chains, modular stack designs, and off-the-shelf, safe and non toxic fast kinetics solvent. Modular scale-up is built into our roadmap: the 1 tpd Paris pilot represents 1/15th of our intended maximum modular size for a single stack. We've advanced from TRL3 to TRL6 in under two years, with our prototype achieving a 56-fold scale-up from our initial lab unit. Integrated heat pumps and heat management allow for operation in cold geographies, such as Canada and the Nordics. The process is fully electric, enabling operational flexibility to modulate according to grid conditions or heat availability.
- Environmental and social safety: Our non-toxic, non-volatile solvent system can be deployed safely even near dense population centers. No hazardous emissions, solvent degradation, or end-of-life disposal risks.
- Deployment velocity: The 1 tpd, 365 tpy pilot will be operational near Paris in 2026. All CO₂ will be mineralized on-site in collaboration with cement partners, producing sustainable, carbon-negative building materials. MRV will be handled through three identified partners to ensure full chain traceability and credit integrity. This pilot will demonstrate the core module for future scale-up - enabling rapid, repeatable deployment at 5,000 tpy and then at the million-ton level.

Alignment with Frontier's innovation priorities

- Durability / permanent storage: Mineralization in cement/concrete waste delivers >1,000-year durability and eliminates the risk of reversal, without the safety or monitoring risks of geological storage.
- Physical footprints: Our absorption kinetics are fast, allowing for a smaller air contactor compared to other technologies. Moreover, the electrochemical technology we use is more efficient than other DAC technologies, resulting in a smaller overall site footprint.
- Cost: FOAK costs <\$350/tCO₂ and <\$150/tCO₂ at the million-ton scale.
- Capacity: Our containerized, modular approach is engineered for rapid, global roll-out using mature supply chains and off-the-shelf equipment derived from other industries.
- Net negativity: Captured CO₂ is permanently mineralized into aggregates. The system is fully electric and can flex with clean grids.
- Additionality: All removals generated by Yama represent new, project-based removals that would not have occurred otherwise. The CO₂ stored in mineralized aggregates is additional to baseline activities and does not overlap with existing emissions reduction obligations. Also Yama DAC can adapt to flexibility electricity need, and can leverage very low grade heat from saline aquifers' brine or data centers heat as a synergy.
- Verifiability: co-located mineralization enables direct, auditable mass balance and on-site sampling - streamlining the MRV process and credit issuance.
- Environmental justice: The technology is safe, non-toxic, and non-volatile, enabling deployment near urban and industrial sites without risk to local communities. Yama does not use any rare materials, only available and scalable materials at solvent and membranes levels. Yama emphasizes local partnerships (e.g., with cement companies near Paris).
- Safety and legality: Yama process uses benign non toxic solvents and avoids hazardous waste streams.

Project details and current status

We are selling carbon credits from our commercial pilot, which is designed to capture 365 tons of CO₂ per year at a TRL6 level. The unit incorporates the largest electrochemical stack we have built, and its air contactor is 1/15th the size of our ultimate commercial unit, thus minimizing technical risk. This pilot will serve as the model for replicable, modular scale-up, enabling the transition from thousands to millions of tons of CO₂ captured annually at ever-lower cost.

We have actively explored robust MRV solutions, with three potential verification partners, and our first credits will be delivered in Q4 2026. The project is already partially de-risked: semi-industrial components are proven in our TRL6 pilot, and our current system has already captured over 50 kg of CO₂ in lab and field tests. Our scale-up pathway is validated by rapid progress (TRL 3/4 to TRL6 in two years), and we have the financial backing of leading climate investors and commercial offtakers including Klarna and Milkywire. Total raised funding is \$4.7 million to date.

Uncertainties and risk management

Remaining uncertainties are in the area of large-scale air contactor performance, further optimization of absorption kinetics, and securing stable, cost-effective supplies of bipolar membranes and non toxic fast kinetics solvents as we scale. Our rapid advance in TRL and the pace of technical improvement underscore our team's capacity to execute and adapt. Also we have secured suppliers for membranes and stacks including chinese manufacturers.

- b. **Project objectives:** What are you trying to build? Discuss location(s) and scale. What is the current cost breakdown, and what needs to happen for your CDR solution to approach Frontier's cost and scale criteria?² What is your approach to quantifying the carbon removed? Please include figures and system schematics and be specific, but concise. 1000-1500 words

Yama's objective is to deploy and scale a new class of modular, non toxic and hybrid electrochemical DAC systems, delivering permanent carbon removal at cost and scale targets that match or surpass Frontier's criteria. Our near-term focus is on operationalizing our Paris-region pilot (365 tpy nameplate), which will demonstrate industrial readiness, cost reduction, and the foundation for multi-kilotonne and eventually megatonne-scale systems. We aim to deliver the first market-validated, cost-disruptive, and replicable DAC platform for the global built environment.

What are we building?

We are building a fully modular DAC 3.0 system that integrates a proprietary hybrid bipolar electro dialysis process with on-site, permanent CO₂ mineralization in recycled cement/concrete waste. Our current commercial pilot unit (TRL6) will have the maximum capacity to capture 365 tonnes CO₂ per year. In practice, we expect to gradually ramp-up in 4 weeks to an average uptime of 76%.

² We're looking for approaches that can reach climate-relevant scale (about 0.5 Gt CDR/year at \$100/ton). We will consider approaches that don't quite meet this bar if they perform well against our other criteria, can enable the removal of hundreds of millions of tons, are otherwise compelling enough to be part of the global portfolio of climate solutions.

This translates into an effective capture of 277 tonnes of CO₂ per year - 224 tonnes net including sequestration - resulting in a total of 449 tonnes of CO₂ net over two years. This will be achieved using the largest electrochemical stack currently available on the market. The air contactor in this system is 1/15th of our target maximum size, minimizing technical risk for future scale-up. The process is fully electric, closed-loop, and designed for both energy and operational efficiency. All captured CO₂ is mineralized on site to create carbon-negative aggregates for sustainable building materials.

Locations and scale

Our first commercial pilot is located near Paris, France, adjacent to a concrete recycling and cement waste handling facility, in partnership with a leading cement player. This location is strategic for several reasons:

- Access to low CO₂ content electricity and potential waste heat source from Data Centers (30°C approx. temperature close to our need)
- Yama team proximity for troubleshooting and optimization.
- Mature local supply (Gennevilliers) chain for recycled aggregates and cementitious waste.
- Regulatory alignment with emerging EU CDR and construction product standards.
- Proximity to MRV - verification partners (e.g. Isometric in London, UK).

After demonstration and validation of the 1 tpd core modular unit, we will replicate and scale up through the following roadmap:

- Q4 2026: Paris pilot operational (1 tpd, 365 tpy).
- 2028: Expansion to a 5,000 tpy demonstrator (using up to 15 modular units).
- 2030 onward: Scale-up to a FOAK (36,500 tpy), with modular replication planned for other European and North American industrial and recycling hubs. The stack, contactor, and process design are optimized for transport, containerization, and local integration.

Current cost breakdown and pathway to Frontier targets

Our cost structure is fully transparent, based on real lab and pilot data, and is protected by IP (2 patents) covering both process and integration.

Current pilot cost at scale:

<\$150/tCO₂ at the million-tonne annualized scale, as demonstrated in our current version (TRL6 data, extrapolated to 1 Mt/year). This includes all Capex (inside battery limits, balance of plant, system integration) and Opex (electricity, membranes, solvents, maintenance, labor, MRV). This is achievable thanks to stack and contactor scale-up, membrane optimization, and supply chain maturity.

For comparison, state-of-the-art thermal DAC systems remain above \$600/tCO₂ at similar scales, and most electrochemical competitors face higher Capex or Opex from exotic or toxic materials with claims around \$300-400/t CO₂.

Key enablers for further cost reduction:

- Scaling up stack technology and contactor sizes while retaining performance.
- Securing long-term supply and cost-down of membranes and non toxic fast kinetics solvents through manufacturing partnerships.
- Optimizing process integration for heat management and electricity use, including coupling with low-carbon grids - so called “flexibility” or industrial waste heat such as DataCenter (approx. 30-35°C).

- Streamlining MRV, labor, and O&M through automation and digital integration.

What needs to happen to reach Frontier's cost and scale criteria?

- Achieve sustained operation of the 1 tpd Paris pilot to confirm TRL6 reliability and validate MRV, crediting, and product durability.
- Leverage modular design for phased scale-up to multi-kilotonne and megatonne annual throughput, with aggressive cost learning through repeated unit deployment.
- Finalize supply agreements and further optimize the procurement of membranes, stack components, and solvents.
- Demonstrate 500h continuous operation (24/7) on our 20 kg CO₂/d prototype.
- Continue solvent and stack innovation to lower both energy use and lifetime replacement rates.
- Demonstrate at the lab the advantage of advanced and emerging bipolar membranes in Yama's DAC process.
- Engage with Frontier-aligned buyers and regulatory stakeholders to secure offtake and support commercial scale financing.

Approach to quantifying carbon removed (MRV)

We are committed to a fully transparent, field-proven MRV process that covers the entire chain from air capture to mineralization. Our approach includes:

- Continuous measurement of inlet and outlet air flows at the contactor, with validated CO₂ concentration readings.
- Direct measurement of solvent loading and CO₂ recovery at each process stage.
- On-site mass balance from stack to mineralization reactor, with sampling of both input waste material and output carbonate aggregates.
- Full digital tracking of operational parameters and material flows, integrated with automated reporting.
- Partnerships with third-party MRV providers (three identified) for independent verification and alignment with standards from Verra, Gold Santard, and Isometric.

This approach enables real-time crediting, robust auditability, and high buyer confidence. The integrated nature of our mineralization pathway allows for direct physical verification; every tonne captured is mineralized on-site and tracked through product delivery.

- c. **Risks:** What are the biggest risks and how will you mitigate those? Include technical, project execution, measurement, reporting and verification (MRV), ecosystem, financial, and any other risks. 500-1000 words

Yama's project is structured to address and minimize risks across technical, execution, MRV, ecosystem, and financial dimensions. Our experience advancing from lab to pilot scale in under two years, our strong IP position, and our collaborations with commercial, MRV, and supply chain partners collectively de-risk the pathway to commercial-scale CDR. Still, key challenges remain. Our mitigation strategies are active and evidence-based.

Technical Risks

- Air Contactor Kinetics: Achieving consistently high CO₂ absorption rates at larger scales is a primary technical challenge. Lab and pilot results are promising, but real-world conditions introduce additional variables such as temperature, humidity, and dust variations.
=> *Mitigation: Ongoing R&D focuses on optimizing non toxic fast kinetics blends and contactor design. We are operating in real conditions on our 20 kg CO₂/d prototype, subject to real conditions variations. We have built a specific intermediate air contactor to operate into the pilot's operating conditions for rapid iterative testing and validation (solvent blending and packing design).*
- Stack performance and durability: While our BPED stacks have demonstrated durability in the lab and first pilot months, extended operation at higher current densities and larger scales may reveal new issues with membrane fouling, scaling, or loss of selectivity.
=> *Mitigation: Continuous monitoring and preventive maintenance are built into the pilot. We are working closely with membrane suppliers for iterative improvements and have included redundancy in key components. We have been benchmarking the membrane to be implemented into our stack according to their durability in Yama's specific process conditions.*
- Microbial growth: Operating in real conditions over extended period of time may lead to microbial growth in our process closed-loops. It may create fouling in the packing of the air contactor or fouling over the membranes of the BPED stacks, leading to loss of performances.
=> *Mitigation: We have been operating in real conditions over 2 months at our prototype, monitoring bacterial growth and process performances. We have mitigating strategies to be implemented such as addition of anti-bacterial agent or cleaning-in-place strategies with acid and/or basic streams on regular basis.*
- Solvent stability: Non toxic fast kinetics solvents offer advantages but may still face unexpected degradation, foaming, or impurity buildup at scale.
=> *Mitigation: Closed-loop solvent management, in-line filtration, and real-time chemical monitoring are implemented. Solvent recovery and replacement costs are tracked closely.*

Project Execution Risks

- Scale-up uncertainties: Scaling from a 1 tpd pilot to multi-kilotonne plants carries risks in integration, supply chain, and cost estimation.
=> *Mitigation: The pilot is designed as a direct modular precursor to commercial units, minimizing unknowns. We use off-the-shelf components, tailored for the CO₂ application, and have begun supply chain negotiations for membranes, stack materials, and solvents.*
- Detailed engineering of the process containerization: Current Yama's prototype has not yet been containerized in order to gain flexibility for potential modifications on what we consider to be a R&D facility. Even though feasibility is demonstrated, the execution of the detailed engineering remains to be done.
=> *Mitigation: Yama's prototype has been operated for several months and gained a solid operational basis to be used as the design basis for the process containerization.*
- Timeline delays: Delays in permitting, equipment delivery, or construction could impact the project's critical path.
=> *Mitigation: The Paris site leverages an existing industrial footprint and regulatory relationships. Early procurement, tight project management, and engagement with experienced EPC partners help reduce risk.*

MRV Risks

- Full-Chain quantification: Achieving high-integrity MRV across DAC and mineralization steps is essential for credit credibility.
=> *Mitigation: The system is designed for continuous, automated measurement of CO₂ flows*

at every stage, with mass balance validated by both process data and aggregate sampling. Three MRV partners are shortlisted for third-party verification and ongoing protocol alignment.

- Evolving MRV standards: The CDR MRV landscape is still developing, with evolving standards for mineralization and hybrid DAC systems.
=> *Mitigation: Active engagement with Verra, Gold Santard (already working with the mineralization tech), and Isometric ensures Yama is positioned to meet or exceed emerging best practices.*

Ecosystem and regulatory risks

- Supply chain resilience: Rapid scaling may be constrained by membrane, solvent, or component availability or cost volatility.
=> *Mitigation: We use mature, non-exotic materials and maintain relationships with multiple vendors. Our solvent is based on specific non toxic fast kinetics chemistry, reducing risk compared to proprietary or hazardous alternatives.*
- Policy and market uncertainty: Shifts in carbon market value, credit acceptance criteria, or policy incentives could impact project economics.
=> *Mitigation: The project is sited in a progressive regulatory environment (France/EU), with multi-year offtake agreements and broad engagement with the voluntary and compliance CDR ecosystem.*

Financial risks

- Fundraising success and business traction: Scaling from pilot to commercial deployment depends on our ability to raise sufficient capital at each phase. Fundraising is directly tied to business traction—successful pilot operation, high-integrity credit pre-sales, and strategic partnerships.
=> *Mitigation: We are engaging actively with climate-focused investors and institutional backers, while leveraging early buyer commitments (Klarna, Milkywire) as proof points. Our modular, milestone-based approach allows us to align fundraising with technical progress and market validation, reducing capital risk at each stage.*
- Cost overruns: Project costs (Capex and Opex) may exceed projections due to unforeseen engineering, integration, or supply chain issues as we scale.
=> *Mitigation: Modular deployment enables phased investment, isolating risk and allowing early identification and correction of cost drivers. We use off-the-shelf and widely available materials to minimize exposure to supply shocks or price volatility.*
- Revenue timing: Delays in credit issuance, verification, or customer acceptance can impact project cash flow, especially at scale.
=> *Mitigation: Pre-sale agreements and milestone-based financing provide a buffer against short-term timing risk. Diversifying our buyer pipeline and aligning MRV with leading standards maximize acceptance and payment reliability.*
- Market acceptance: Long-term financial viability requires sustained market demand and robust credit acceptance in both voluntary and compliance markets.
=> *Mitigation: We prioritize MRV transparency, active stakeholder engagement, and close alignment with evolving standards to ensure credits are recognized and valued by key buyers.*
- BPED market scale-up: membrane and stack costs are expected to fall significantly, following standard learning rate effects. Current projections indicate a 3.5-4x reduction from today's prices.
=> *Mitigation: To capture these cost reductions and secure supply, Yama is reinforcing partnerships with leading membrane and stack suppliers in Europe, the USA, and Asia. If*

required, we are prepared to establish a dedicated manufacturing facility through a joint venture with manufacturers and polymer suppliers.

- d. **Proposed offer to Frontier:** Please list proposed CDR volume, delivery timeline and price below. If you are selected for a Frontier prepurchase, this table will form the basis of contract discussions.

Proposed CDR over the project lifetime (tons) <i>(should be net volume after taking into account the uncertainty discount proposed in 5c)</i>	255 tons CO ₂ net removed
Delivery window <i>(at what point should Frontier consider your contract complete? Should match 2f)</i>	Q1 2027 - Q4 2028
Levelized cost (\$/ton CO ₂) <i>(This is the cost per ton for the project tonnage described above, and should match 6d)</i>	\$3,832/tCO ₂ net removed (including the cost of capital, sequestration, MRV)
Levelized price (\$/ton CO ₂) ³ <i>(This is the price per ton of your offer to us for the tonnage described above)</i>	\$1700/tCO ₂ net removed

³ This does not need to exactly match the cost calculated for “This Project” in the TEA spreadsheet (e.g., it’s expected to include a margin and reflect reductions from co-product revenue if applicable).